Experiment No 8 - Configuring MetaMask with a DApp (TRUST)

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Aim

To configure a cryptocurrency wallet using MetaMask and integrate it with the TRUST escrow DApp, so the wallet can connect to a local Hardhat blockchain, approve token spending, lock a payment in a smart-contract escrow and show the result on-chain.

Theory

MetaMask is a browser wallet that holds blockchain accounts and signs transactions. A decentralized application (DApp) can only ask the wallet for access and send it requests; every transaction still needs the user's approval in MetaMask. In the connection request used here, the site is allowed to see addresses, balances and activity and to send transaction requests, but it cannot move funds without permission.

TRUST runs against a local Hardhat blockchain (chain ID 31337) instead of a public network, so test ETH pays for gas and a Mock USDC token (ERC-20, 6 decimals) is the payment currency. Moving ERC-20 tokens into a contract takes two approvals:

1. **Spending cap (approve).** The DApp asks Mock USDC to let the escrow contract spend up to a set amount from the user's account.
2. **Escrow transaction.** The DApp calls the TrustEscrow contract, which pulls the approved amount and locks it as an escrow.

Each escrow is stored on-chain with a status: 1 = Funded, 2 = Released, 3 = Refunded. The advertiser releases the payment once the creator's post proves the agreed reach, and can refund if the work is not delivered after 7 days. A local blockchain explorer page reads this state directly from the Hardhat JSON-RPC node, so the result of each MetaMask approval can be checked on-chain.

Steps

1. Add the Hardhat Local network (chain ID 31337) to MetaMask and open the TRUST DApp at http://localhost:3000/advertiser.
2. In the wallet selector in the header, choose MetaMask instead of the built-in Advertiser wallet.
3. Review the MetaMask connection request for localhost, check the requested permissions, and select Connect for Account 1 (Fig. 1).
4. If MetaMask is locked when a request arrives, enter the MetaMask password and select Unlock (Fig. 2).
5. Open the Creators list, select Hire on @mr_zoseidon, and enter the amount (2400 USDC) and target views (1). Add an optional brief, then submit. The dialog shows "Working... (step 1 of 2)".
6. Step 1: review the spending-cap request in MetaMask. Check the cap (2,400), the spender (the escrow contract), the network (Hardhat Local) and the requesting site, then select Confirm (Fig. 3). MetaMask marks the site with an HTTP warning because localhost is not served over HTTPS; this is expected for local development.
7. Step 2: review the transaction request sent to the escrow contract, check the network fee, and select Confirm (Fig. 4).
8. After the creator posts, open the submitted post in the DApp. TRUST reads the public counters and shows a verification report. Choose Verify & release, Refund or Dispute (Fig. 5).

9. Open the Explorer page and check the escrow list, the vault balance and the latest block to confirm the on-chain state (Fig. 6).

Output

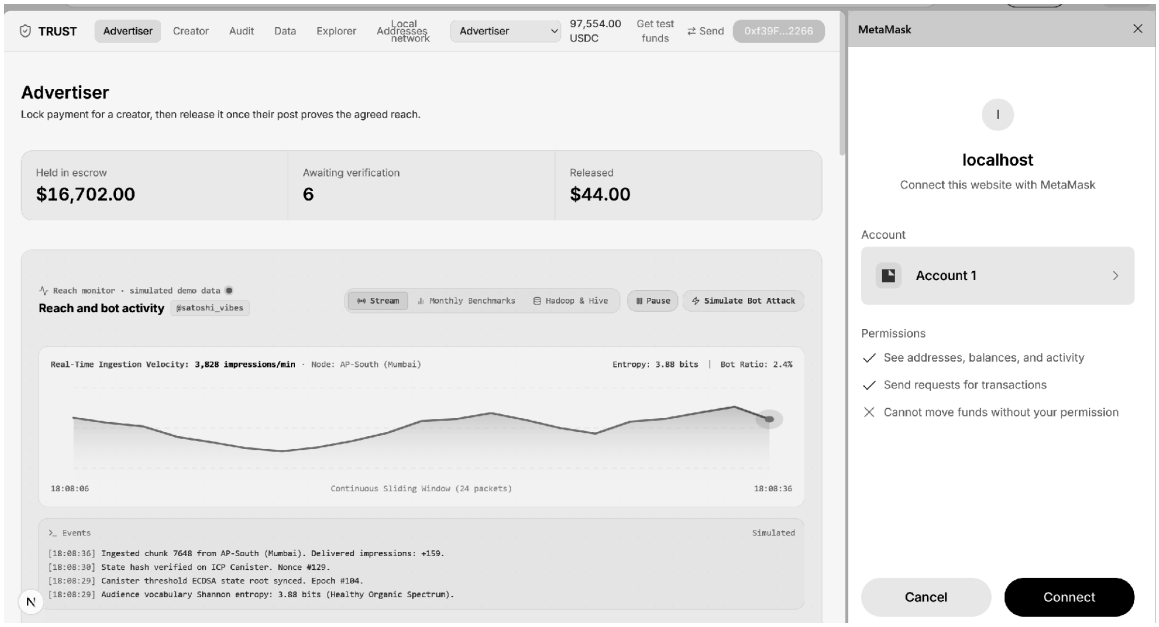


Fig. 1. MetaMask connection request for the TRUST DApp (localhost), with the requested permissions.

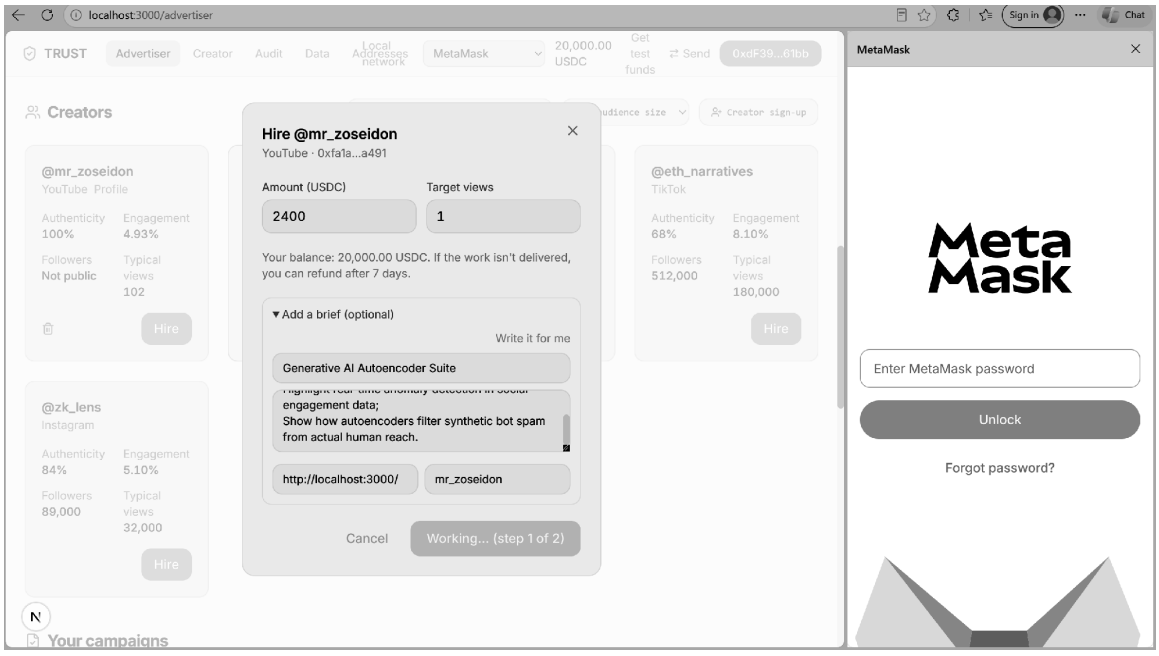


Fig. 2. MetaMask asking for the wallet password during the Hire flow.

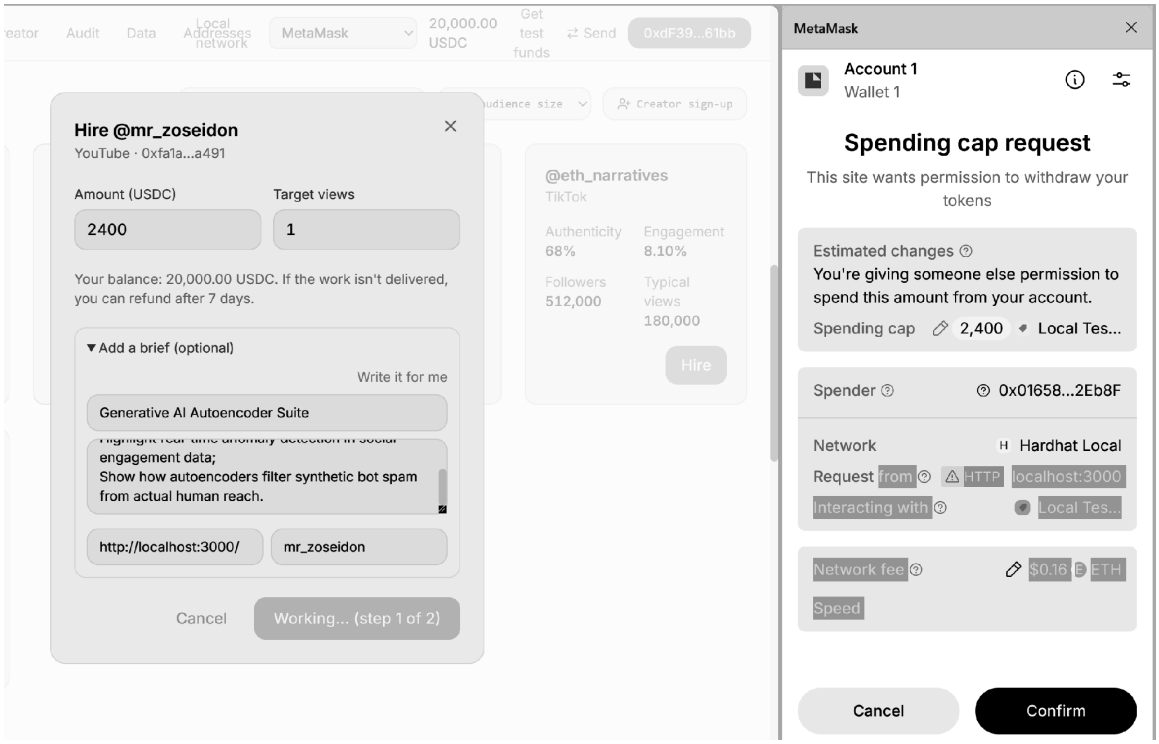


Fig. 3. Spending-cap request (step 1 of 2): 2,400 Mock USDC for the escrow contract, on Hardhat Local.

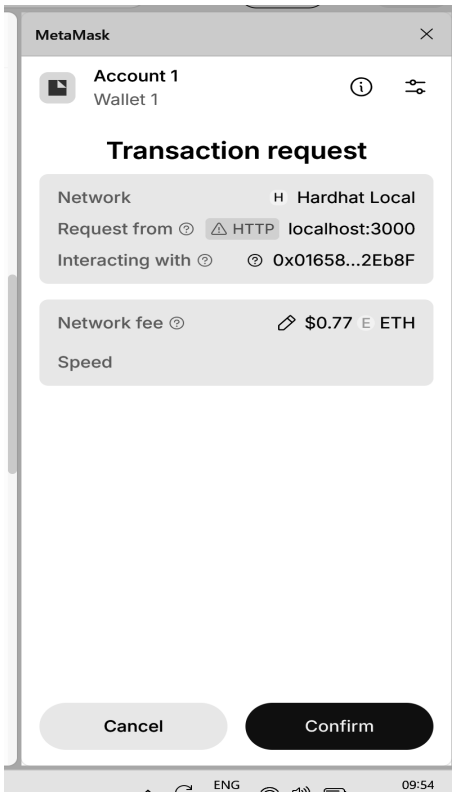


Fig. 4. Transaction request to the escrow contract on Hardhat Local (step 2 of 2).

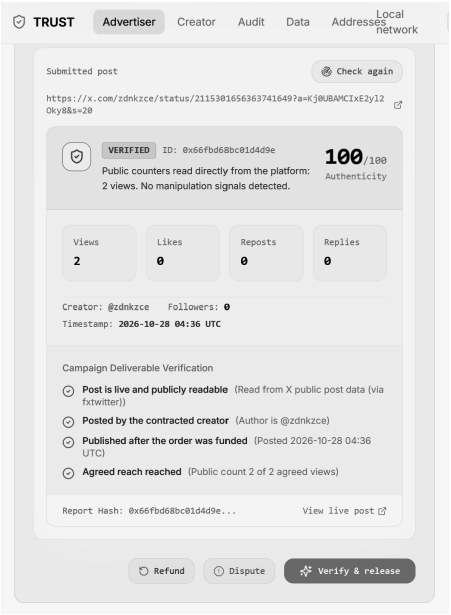


Fig. 5. Verification report for a creator's submitted post, with the Refund, Dispute and Verify & release actions.

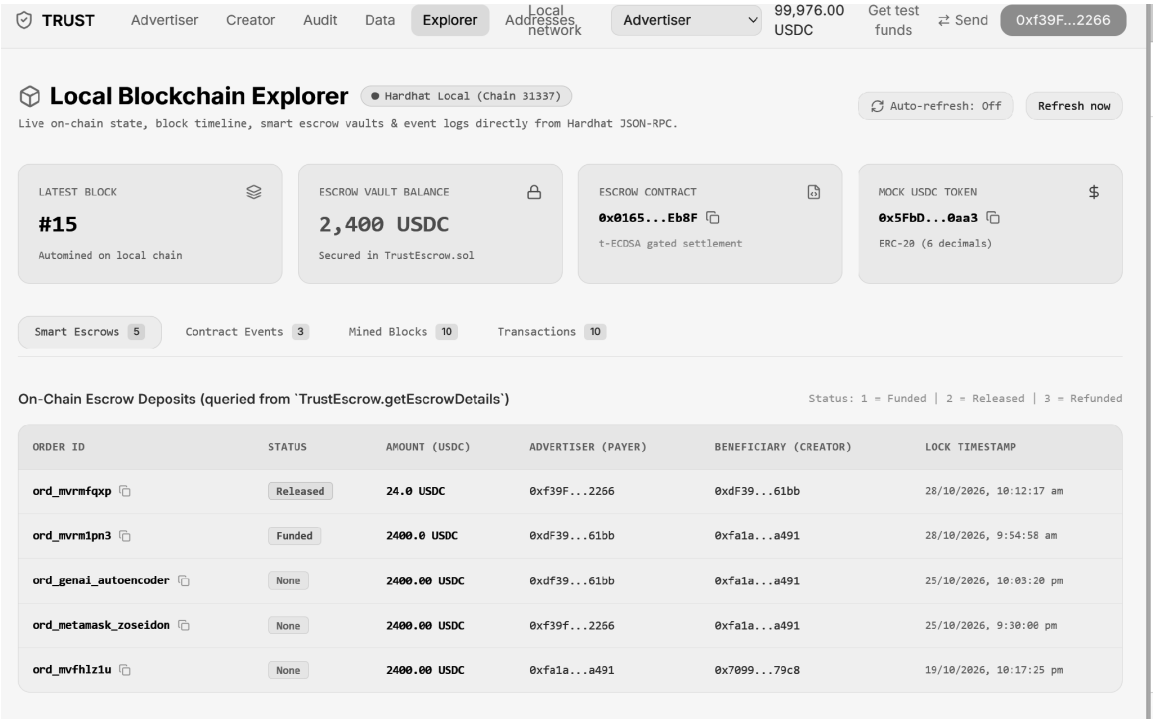


Fig. 6. Local blockchain explorer: latest block #15, 2,400 USDC held in the escrow vault, and the on-chain list of escrows with their status.

Conclusion

MetaMask was connected to the TRUST DApp on the Hardhat Local network, and each request it sent was reviewed and approved in the wallet: the connection permissions, the 2,400 USDC spending cap, and the transaction to the escrow contract. The explorer then confirmed the result on-chain: a Funded escrow of 2,400 USDC matching the 2,400 USDC vault balance, and a 24 USDC escrow marked Released.